

A Composition Theorem for Pipeline Deletion

Standalone preview — drop-in section for the ADCU manuscript

1 A Composition Theorem for Pipeline Deletion

The experiments show repeatedly that deleting a record from each component in isolation does not delete it from the deployed system: an index can be purged while a summary survives, or an adapter can be unlearned while a synthetic derivative is still retrieved. This section makes that observation precise. We define behavioral deletion against a counterfactual reference system, prove that per-component deletion guarantees *compose* up to an explicit derivative-coupling term, and prove that this coupling term is *necessary*: even perfect, independently certified component-level deletion does not imply pipeline-level deletion. The theorem provides the formal justification for treating deletion verification as a provenance-closed, multi-channel behavioral audit, which is exactly what ADCU operationalizes.

1.1 Setup and Counterfactual Reference

Fix a deletion target z . A foundation-model pipeline produces, for a query q , a distribution over answers. We write the pipeline’s answer law as $P_S(\cdot \mid q)$ for a system S . A system is characterized by a parameter state θ (base weights and adapters) and an external knowledge state K (vector index, lexical index, reranker state, caches, summaries, and any synthetic-derivative store). We consider queries drawn from a declared probe class $\mathcal{B}$ and measure deviation between answer laws by total variation, $\text{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$.

We compare three systems built from the same architecture:

- S_0 , the pre-deletion system (θ_0, K_0) ;
- S^- , the deployed post-deletion system (θ^-, K^-) obtained by applying a deletion operation to S_0 ;
- S^* , the *counterfactual reference* (θ^*, K^*) , i.e. the pipeline that would have been built had z *never been ingested*. S^* is the gold standard inherited from exact retraining and certified removal, lifted here from a single model to the whole pipeline.

Definition 1 (Behavioral residual influence). The behavioral residual influence of z in the post-deletion system, relative to the probe class $\mathcal{B}$, is

$$\rho_{\mathcal{B}}(S^-) = \sup_{q \in \mathcal{B}} \text{TV}(P_{S^-}(\cdot \mid q), P_{S^*}(\cdot \mid q)).$$

We say S^- achieves ε -behavioral deletion on $\mathcal{B}$ if $\rho_{\mathcal{B}}(S^-) \leq \varepsilon$.

Definition 1 is the property an auditor actually cares about: it asks whether the deployed system is behaviorally indistinguishable, over the probe class, from a system that never saw z . It is strictly stronger than checking that artifacts labeled with z are gone, a distinction we now name.

Definition 2 (Syntactic vs. behavioral deletion). A deletion operation is *syntactic* if it removes exactly the artifacts explicitly tagged with the provenance identifier of z . It is *behavioral* if the resulting system attains ε -behavioral deletion for small ε . Syntactic deletion is observable from metadata; behavioral deletion is a property of the answer law and is what Definition 1 measures.

1.2 Pipeline Decomposition

We model the canonical retrieval-plus-generation pipeline as two stages with the parameter state entering generation as a side input. Given q and knowledge state K , a retrieval stage produces a (random) context X with law $\mathcal{R}(\cdot \mid q, K)$. A generation stage produces the answer with law $\mathcal{G}(\cdot \mid q, X, \theta)$. The pipeline answer law is the mixture

$$P_S(\cdot \mid q) = \mathbb{E}_{X \sim \mathcal{R}(\cdot \mid q, K)} [\mathcal{G}(\cdot \mid q, X, \theta)]. \quad (1)$$

Deeper pipelines (reranking, caching, summarization) are obtained by further composing stages; the serial generalization is Theorem 2.

We assume the deletion operation is, or can be analyzed relative to, a *provenance-closed* idealization. Let $\mathcal{D}(z)$ denote the transitive closure of z in the provenance graph: every chunk, embedding, summary, cache entry, synthetic instruction/QA example, and fine-tuning record causally derived from z or from a member of $\mathcal{D}(z)$. Let $S^\diamond = (\theta^\diamond, K^\diamond)$ be the system obtained from S^- by *additionally* removing all of $\mathcal{D}(z)$ from the knowledge state and excluding it from the data that produced the parameters. $S^\diamond$ is the system S^- “should have been” had deletion followed lineage rather than labels.

1.3 Assumptions

Assumption 1 (Parametric behavioral deletion). Uniformly over the probe class and over contexts,

$$\sup_{q \in \mathcal{B}, x} \text{TV}(\mathcal{G}(\cdot \mid q, x, \theta^\diamond), \mathcal{G}(\cdot \mid q, x, \theta^\star)) \leq \varepsilon_\theta.$$

Assumption 2 (Knowledge-state deletion). For a retrieval discrepancy $d_{\mathcal{R}}$,

$$\sup_{q \in \mathcal{B}} d_{\mathcal{R}}(\mathcal{R}(\cdot \mid q, K^\diamond), \mathcal{R}(\cdot \mid q, K^\star)) \leq \varepsilon_K.$$

Assumption 3 (Bounded context sensitivity). The generation stage is L -Lipschitz from the retrieval discrepancy to answer total variation: for any context laws α, β and any fixed (q, θ) ,

$$\text{TV}(\mathbb{E}_{X \sim \alpha} \mathcal{G}(\cdot \mid q, X, \theta), \mathbb{E}_{X \sim \beta} \mathcal{G}(\cdot \mid q, X, \theta)) \leq L d_{\mathcal{R}}(\alpha, \beta).$$

Assumptions 1–2 say the parametric and knowledge components are individually deletion-certified *against the counterfactual references on the provenance-closed problem*. Assumption 3 bounds how strongly a change in what is retrieved can change what is answered. When $d_{\mathcal{R}}$ is itself total variation over context laws, $L \leq 1$ by the data-processing inequality (mixing is a channel), and (1) gives no amplification; $L > 1$ captures the practical case where ε_K is reported in a weaker retrieval metric (e.g. retrieved-identifier overlap) than the answer divergence it induces.

1.4 Composition

Lemma 1 (Hybrid bound). *For any q ,*

$$\text{TV}(P_{S^\diamond}(\cdot \mid q), P_{S^\star}(\cdot \mid q)) \leq \varepsilon_\theta + L \varepsilon_K.$$

Proof. Insert the hybrid system $(\theta^\star, K^\diamond)$ and apply the triangle inequality:

$$\begin{aligned} \text{TV}(P_{(\theta^\diamond, K^\diamond)}, P_{(\theta^\star, K^\star)}) &\leq \text{TV}(P_{(\theta^\diamond, K^\diamond)}, P_{(\theta^\star, K^\diamond)}) \\ &\quad + \text{TV}(P_{(\theta^\star, K^\diamond)}, P_{(\theta^\star, K^\star)}). \end{aligned}$$

For the first term the knowledge state is fixed at $K^\diamond$, so by (1) and convexity of total variation under mixtures,

$$\text{TV}(P_{(\theta^\diamond, K^\diamond)}, P_{(\theta^\star, K^\diamond)}) \leq \mathbb{E}_X \text{TV}(\mathcal{G}(\cdot | q, X, \theta^\diamond), \mathcal{G}(\cdot | q, X, \theta^\star)) \leq \varepsilon_\theta,$$

using Assumption 1. For the second term the parameters are fixed at $\theta^\star$ and only the retrieval law differs, so by Assumption 3 and then Assumption 2,

$$\text{TV}(P_{(\theta^\star, K^\diamond)}, P_{(\theta^\star, K^\star)}) \leq L d_{\mathcal{R}}(\mathcal{R}(\cdot | q, K^\diamond), \mathcal{R}(\cdot | q, K^\star)) \leq L \varepsilon_K.$$

Summing gives the claim. $\square$

We now isolate the part of the deviation that per-component guarantees *cannot* reach: the orphan derivatives present in S^- but absent from the provenance-closed $S^\diamond$.

Definition 3 (Derivative-coupling term).

$$\gamma_z = \sup_{q \in \mathcal{B}} \text{TV}(P_{S^-}(\cdot | q), P_{S^\diamond}(\cdot | q)).$$

We say the deletion operation achieves *provenance closure* if $S^- = S^\diamond$, equivalently if every artifact in $\mathcal{D}(z)$ was removed.

Theorem 1 (Pipeline deletion composition). *Under Assumptions 1–3,*

$$\boxed{\rho_{\mathcal{B}}(S^-) \leq \varepsilon_\theta + L \varepsilon_K + \gamma_z.}$$

Moreover $\gamma_z = 0$ if and only if the deletion operation achieves provenance closure, in which case $\rho_{\mathcal{B}}(S^-) \leq \varepsilon_\theta + L \varepsilon_K$.

Proof. By the triangle inequality, for each q ,

$$\text{TV}(P_{S^-}, P_{S^\star}) \leq \text{TV}(P_{S^-}, P_{S^\diamond}) + \text{TV}(P_{S^\diamond}, P_{S^\star}).$$

Taking $\sup_{q \in \mathcal{B}}$, the first supremum is γ_z by Definition 3 and the second is bounded by $\varepsilon_\theta + L \varepsilon_K$ by Lemma 1. If provenance closure holds then $S^- = S^\diamond$, so $\gamma_z = 0$; conversely $\gamma_z = 0$ forces $P_{S^-} = P_{S^\diamond}$ on $\mathcal{B}$, i.e. the retained derivatives exert no behavioral influence over the probe class, which is provenance closure in behavioral form. $\square$

The next result shows the coupling term is not an artifact of loose analysis: component-wise deletion, *even when perfect*, is insufficient.

Proposition 1 (Necessity of provenance closure). *For every $\delta \in (0, 1)$ there is a pipeline and a deletion operation with $\varepsilon_\theta = \varepsilon_K = 0$ —each component individually certified against its counterfactual reference and free of any artifact labeled z —yet $\rho_{\mathcal{B}}(S^-) \geq 1 - \delta$. Hence no bound of the form $\rho_{\mathcal{B}}(S^-) \leq f(\varepsilon_\theta, \varepsilon_K)$ can hold without the coupling term γ_z .*

Proof. Let the base model never be trained on z , so $\theta^- = \theta^\diamond = \theta^\star$ and $\varepsilon_\theta = 0$. During ingestion a summarization component writes a derivative $s(z) \in \mathcal{D}(z)$ —a paraphrased summary of z ’s protected fact—into the index under a *fresh* artifact identifier that is not tagged with z . Syntactic deletion removes the raw chunk of z but, having no label linking $s(z)$ to z , leaves $s(z)$ in K^- . Restrict the comparison to direct artifacts of z : relative to those, the retrieval laws of K^- and $K^\star$ coincide, so a label-based audit reports $\varepsilon_K = 0$.

Now take a probe $q \in \mathcal{B}$ asking for the protected fact. Under S^- the retriever returns $s(z)$ with probability one and the generator answers the fact; under S^* neither z nor $s(z)$ was ever ingested, so the system returns “insufficient information.” These answer laws are almost disjoint, giving $\text{TV}(P_{S^-}(\cdot | q), P_{S^*}(\cdot | q)) \geq 1 - \delta$ for any target δ by making the two responses distinguishable with probability $\geq 1 - \delta$. Therefore $\rho_{\mathcal{B}}(S^-) \geq 1 - \delta$ while $\varepsilon_{\theta} = \varepsilon_K = 0$; the entire deviation is carried by γ_z , the orphan derivative $s(z)$. $\square$

Proposition 1 is the formal counterpart of the *synthetic derivative retained* and *shadow-copy* failures in our experiments, where each component looks clean by its own check yet the pipeline still answers the deleted fact. It explains why those rows carry near-maximal risk even though no artifact labeled with the target remains.

1.5 Serial Pipelines

Theorem 2 (Multi-stage composition). *Consider a serial pipeline $C_m \circ \dots \circ C_1$ in which stage C_i has deletion error ε_i against its counterfactual reference (on the provenance-closed problem) and every downstream stage C_k , $k > i$, is L_k -Lipschitz in the sense of Assumption 3. Then*

$$\rho_{\mathcal{B}}(S^-) \leq \sum_{i=1}^m \left(\prod_{k=i+1}^m L_k \right) \varepsilon_i + \gamma_z.$$

Proof. Form the m hybrids that switch stages from the reference to the deployed version one at a time, holding all others fixed. Telescoping the triangle inequality, the deviation introduced when stage C_i is switched is at most ε_i , and it is transported to the output through the downstream stages, each contracting (or amplifying) it by its Lipschitz factor, contributing $(\prod_{k>i} L_k) \varepsilon_i$. Summation over i bounds the provenance-closed deviation $\text{TV}(P_{S^\circ}, P_{S^*})$; adding γ_z as in Theorem 1 gives the result. $\square$

1.6 Consequences for Auditing

Corollary 1 (Audit sufficiency). *To certify ε -behavioral deletion it suffices to (i) establish provenance closure, driving $\gamma_z \rightarrow 0$, and (ii) bound each component error so that the propagated sum in Theorem 2 is at most ε . Conversely, by Proposition 1, an audit that omits either ingredient cannot certify deletion.*

Corollary 1 maps onto ADCU term by term. Building the provenance graph and expanding deletion targets to $\mathcal{D}(z)$ is the operational route to $\gamma_z \rightarrow 0$; the residual-dependence channels estimate the component and behavioral errors that survive closure (direct, paraphrase, and extraction probes estimate the answer-side parametric and derivative error; retrieval-dependence probes estimate the retrieval-mediated term $L\varepsilon_K$; watermark probes detect labeled-artifact survival; the retrieval-off counterfactual isolates which path dominates). The Hoeffding upper confidence bound of Section 4 is then the finite-sample, high-probability estimate of $\rho_{\mathcal{B}}(S^-)$ that Theorem 1 bounds in expectation: a pass certifies that the estimated propagated error plus coupling term lies below tolerance, under the declared probe class.

Relation to differential-privacy composition. Theorem 2 is structurally a composition-across-data-flow result—deletion budgets add (sensitivity-weighted) along the pipeline—but it differs from differential-privacy composition in an essential way: it carries the non-additive term γ_z . Standard privacy composition has no analogue of γ_z because it does not model derived artifacts

that re-enter the pipeline. The coupling term is precisely the contribution of derivative re-ingestion, and it is why deletion in compositional foundation-model systems requires lineage tracking rather than per-component accounting alone.

Assumptions and scope. The guarantee is conditional on the declared probe class $\mathcal{B}$ (it bounds residual influence over $\mathcal{B}$, not over all conceivable prompts), on access to the counterfactual references for defining $\varepsilon_\theta, \varepsilon_K$ (in practice estimated, e.g. via a retrain-without-record reference on a sampled slice), and on stable system behavior during the audit. These are the same operating assumptions used by the empirical metric, now made explicit at the level of the composition they justify.